

SOFIA KIRSANOVA

PhD Student in Computer Science, University of Minnesota Twin Cities

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EDUCATION

Ph.D. in Computer Science, University of Minnesota Twin Cities Knowledge Computing Lab; Advisor: Prof. Yao-Yi Chiang Research: computer vision, document image analysis, multimodal deep learning, geospatial AI	2024-Present
M.Sc. in Computer Science, University of Minnesota Twin Cities Advisor: Prof. Yao-Yi Chiang	2024-2026
B.Sc. in Applied Mathematics and Computer Science, Lomonosov Moscow State University Thesis: Process Analytics Methods for Anomaly Detection Tasks in System Log Data	2019-2023

RESEARCH AREA

Computer Vision | Vision Transformers | Multimodal Models | Geospatial Deep Learning | Layout Understanding | Automated Digitization | Line Extraction & Topological Reconstruction

RESEARCH EXPERIENCE

Graduate Researcher, Knowledge Computing Lab, University of Minnesota, USA <i>IMOLA (NSF). Topology-Preserving Road Network Extraction from Historical Maps (ongoing)</i> - Develop a topology-preserving road network extraction method combining graph-based line detection, segmentation guidance, and node-type classification for historical topographic maps. - Build algorithms for topology reconstruction, converting raw line predictions into routable graph structures with node and edge classification and connectivity correction. - Design evaluation pipelines for large heterogeneous map collections, including patch difficulty modeling, structured error attribution, and cross-region generalization. <i>CriticalMAAS (DARPA). Legend Detection & Map Layout Analysis (completed)</i> - Developed deep-learning methods for detecting legend symbols and text regions and linking them into structured legend items across diverse cartographic styles. - Combined fine-tuned LayoutLMv3 with GPT-4o structured prompting for layout-aware legend parsing in the DIGMAPPER map-digitization pipeline. - Achieved 96% symbol-detection F1, 97.8% text-detection F1, and 97.1% symbol-description linking accuracy.	2024-Present
Undergraduate Researcher, Lomonosov Moscow State University, Moscow, Russia <i>Methods for Detecting Critical Events in System Log Data</i> - Developed preprocessing, semantic extraction, and CNN-based anomaly detection methods for unstructured system logs. <i>Monte Carlo Algorithms for Connect6</i> - Compared Monte Carlo and heuristic algorithms for stochastic gameplay modeling.	2019-2023

PUBLICATIONS

1. Sofia Kirsanova, Zekun Li, Stefan Leyk, Craig A. Knoblock, Yao-Yi Chiang. TRASA: Topology-Preserving Road Network Extraction from Historical Topographic Maps. Manuscript in preparation, 2026.

2. Jihun Pyo, Yuxi Jiao, Daehan Jung, Zekun Li, Leeje Jang, Sofia Kirsanova, Jina Kim, Yijun Lin, Qian Liu, Jie Xie, Hooman Askari, Ning Xu, Muhao Chen, Yao-Yi Chiang. FRIEDA: Benchmarking Multi-Step Cartographic Reasoning in Vision-Language Models. ICLR, 2026.

3. Weiwei Duan, Michael Gerlek, Steven Minton, Craig Knoblock, Fandel Lin, Theresa Chen, Leeje Jang, Sofia Kirsanova, Zekun Li, Yijun Lin, Yao-Yi Chiang. DIGMAPPER: A Modular System for Automated Geologic Map Digitization. ACM SIGSPATIAL, 2025.

4. Sofia Kirsanova, Weiwei Duan, Yao-Yi Chiang. Detecting Legend Items on Historical Maps Using GPT-4o with In-Context Learning. ACM SIGSPATIAL Workshops, 2025.

5. Haoji Hu, Jina Kim, Jinwei Zhou, Sofia Kirsanova, Janghyeon Lee, Yao-Yi Chiang. Context-Aware Trajectory Anomaly Detection. ACM SIGSPATIAL Workshops, 2024.

6. Oleg E. Gorokhov, Sofia Kirsanova. Detection of Anomalies in System Log Data Using Process Analytics Methods. Lomonosov Readings, 2023.

7. Oleg E. Gorokhov, Mikhail I. Petrovsky, Igor V. Mashechkin, Sofia Kirsanova. Methods for Detecting Critical Events in System Log Data. Tikhonov Readings, 2022.

8. Sofia Kirsanova. Evaluation and Comparison of Algorithms for the Connect6 Game with Monte Carlo Method. Moscow Polytechnic University Scientific Fair, 2020.

CONFERENCE PRESENTATIONS

- TRASA: Topology-Preserving Road Network Extraction from Historical Topographic Maps. Frontiers of Generative AI & Science Workshop, University of Minnesota, Aug 2026.
- Detecting Legend Items on Historical Maps. MN GIS/LIS Consortium Conference, Duluth, MN, Oct 2025.
- Detecting Legend Items on Historical Maps. Big Ten GIS Conference (BTAA GIN), Apr 2025.

SELECTED SERVICE

External Reviewer, ACM SIGSPATIAL 2026 Applications Track, 2026.

PROFESSIONAL EXPERIENCE

Machine Learning Researcher, Rotec Digital Solutions, Moscow, Russia	Sep 2023-Apr 2024
- Developed anomaly-detection models for industrial sensor and time-series data and integrated them into monitoring pipelines using PyTorch, scikit-learn, PostgreSQL, ClickHouse, MLflow, DVC, Airflow, and Optuna.	
Full-stack Java Developer, MOLNET, Moscow, Russia	Apr 2022-Apr 2023
- Implemented Java/Spring backend systems and React/TypeScript web interfaces; used PostgreSQL, Docker, Git, and Linux.	
Freelance Writer, JetBrains Academy (remote)	Jan 2022-Mar 2023
- Authored educational modules on probability, statistics, data structures, and algorithms for online CS courses.	

SKILLS

Core Research:	Computer Vision, Document Image Analysis, Layout Understanding, Geospatial Deep Learning, Multimodal Models, Vision Transformers, OCR & LLM-based Extraction
Machine Learning:	PyTorch, scikit-learn, CNNs, transformer architectures, training/evaluation pipelines, dataset curation, experiment tracking
Geospatial & Data Tools:	QGIS, PostGIS, PostgreSQL, geospatial data processing, map digitization workflows
Programming:	Python, C, C++, Java, SQL, Bash, JavaScript/TypeScript
Languages:	Russian (native), English (fluent), German (intermediate)